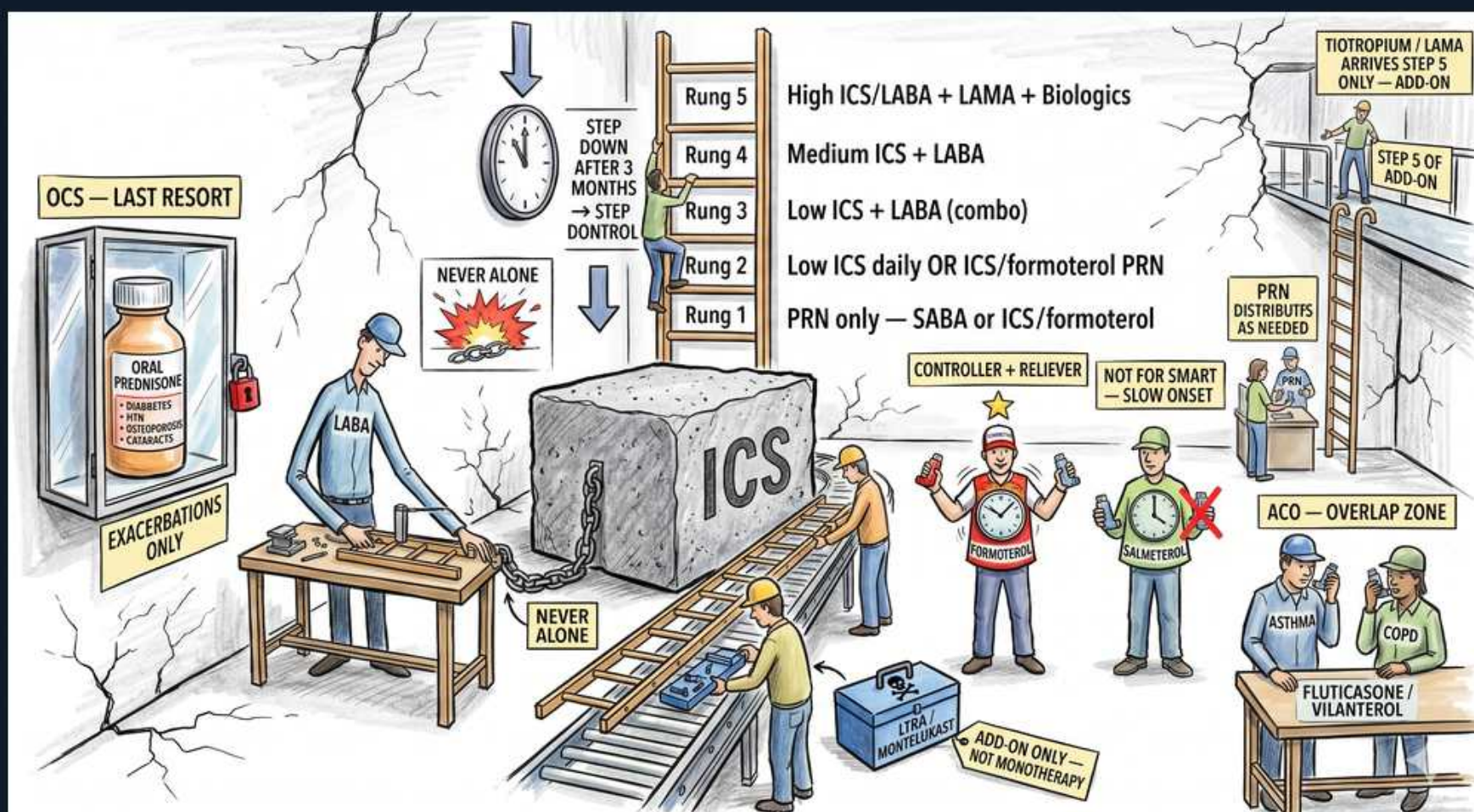


# Asthma — Chronic Step-Up Therapy Ladder





## 1. Step-down after 3 months

### WHAT YOU SEE

A clock with 'STEP DOWN after 3 months → step control'.

### WHAT IT ENCODES

Once asthma is well-controlled for ~3 months, step DOWN to the lowest effective dose that maintains control, reassessing every 3 months. Control is judged over the prior 4 weeks: daytime symptoms  $\leq 2$ /week, no nocturnal waking, reliever  $\leq 2$ /week, no activity limitation — all four = well-controlled. Before stepping UP, always check inhaler technique, adherence, and trigger exposure first.

### BOARD TRAP

Before escalating therapy for 'uncontrolled' asthma, exclude the real culprits — poor inhaler technique, non-adherence, ongoing allergen/smoke exposure, and wrong diagnosis — rather than just adding drugs.

## 2. Step ladder rungs 1-5

### WHAT YOU SEE

A ladder: Rung1 PRN ICS-formoterol, Rung2 low ICS, Rung3 low ICS+LABA, Rung4 medium ICS+LABA, Rung5 high ICS/LABA+LAMA+biologics.

### WHAT IT ENCODES

GINA stepwise management (track 1, preferred): Step 1-2 = as-needed low-dose ICS-formoterol; Step 3 = low-dose maintenance ICS-formoterol; Step 4 = medium-dose ICS-formoterol (all as MART); Step 5 = high-dose ICS-LABA, add LAMA and refer for phenotype-based biologics. The controller (ICS) is introduced early and intensified to control; SABA-only is no longer a recommended track.

### BOARD TRAP

There is no asthma step that is reliever-only with no anti-inflammatory — even mildest asthma gets ICS (as ICS-formoterol PRN), because SABA monotherapy increases exacerbation and death risk.

## 3. PRN only = SABA or ICS-formoterol

### WHAT YOU SEE

Rung 1 reads 'PRN only — SABA or ICS/formoterol'.

### WHAT IT ENCODES

For mild/intermittent asthma, GINA preferred reliever is as-needed low-dose ICS-formoterol (anti-inflammatory reliever, AIR) — it treats inflammation with each use and cuts severe exacerbations versus SABA-only. SABA alone is now only an alternative track requiring concurrent ICS whenever the reliever is taken.

### BOARD TRAP

SABA monotherapy is no longer preferred even in mild asthma — it relieves bronchospasm but does nothing for the underlying inflammation and over-reliance ( $>1$  canister/year) predicts death.

## 4. Tiotropium / LAMA arrives Step 5

### WHAT YOU SEE

A worker carrying tiotropium up to 'LAMA arrives step 5 only — add-on'.

### WHAT IT ENCODES

Tiotropium (a LAMA, via soft-mist inhaler) is an ADD-ON at Step 4-5 for patients still uncontrolled on ICS-LABA — it adds bronchodilation and lengthens time to exacerbation. It is never a stand-alone asthma controller.

### BOARD TRAP

LAMA is add-on therapy in asthma layered onto ICS — never the sole controller and never used without an ICS (unlike COPD, where LAMA monotherapy is appropriate).

## 5. OCS — last resort

### WHAT YOU SEE

A locked cabinet 'OCS — last resort' with oral prednisone and side-effect tags.

### WHAT IT ENCODES

Maintenance oral corticosteroids are a last resort for severe refractory asthma because of cumulative systemic toxicity — diabetes, osteoporosis/fractures, cataracts, adrenal suppression, weight gain, infection. Modern biologics are specifically used to spare or eliminate chronic OCS in eosinophilic/OCS-dependent patients.

### BOARD TRAP

Avoid chronic systemic steroids for maintenance when inhaled options or biologics exist — an OCS-dependent asthmatic is an indication to add a biologic, not to keep escalating prednisone.

## 6. ICS cornerstone block

### WHAT YOU SEE

A massive concrete block labeled 'ICS' being moved by workers.

### WHAT IT ENCODES

Inhaled corticosteroids are the foundational anti-inflammatory controller for all persistent asthma — they suppress eosinophilic airway inflammation, improve control, and are the only controller class proven to reduce asthma mortality. Local side effects (oral candidiasis, dysphonia) are minimized by spacer use and rinsing the mouth after dosing.

## 7. LABA 'never alone'

### WHAT YOU SEE

A worker labeled 'LABA' with a chain and 'NEVER ALONE' / explosion warning.

### WHAT IT ENCODES

A LABA must ALWAYS be paired with an ICS in asthma. LABA monotherapy carries an FDA black-box warning for increased asthma-related death — without the steroid, the LABA masks worsening inflammation while bronchodilating, leading to fatal exacerbations. Fixed-dose ICS-LABA combination inhalers exist precisely to prevent accidental LABA-alone use.

### BOARD TRAP

LABA monotherapy in asthma is dangerous (black-box, increased death) — always combine with ICS; this is the most heavily tested asthma-pharmacology point.



## 8. Controller + reliever combo

### WHAT YOU SEE

A worker holding both a green and red inhaler 'Controller + reliever'.

### WHAT IT ENCODES

A single ICS-formoterol inhaler can serve as BOTH daily maintenance AND as-needed reliever — the MART/SMART regimen (Maintenance And Reliever Therapy). One inhaler simplifies the regimen, improves adherence, and delivers extra anti-inflammatory dosing exactly when symptoms flare.

### BOARD TRAP

MART only works with ICS-FORMOTEROL — you cannot do MART with an ICS-salmeterol inhaler because salmeterol is too slow for rescue; mixing this up is a board trap.

## 9. Formoterol — fast onset

### WHAT YOU SEE

An inhaler 'FORMOTEROL' marked usable as reliever.

### WHAT IT ENCODES

Formoterol is a LABA with a SABA-like rapid onset (~1-3 min) and long duration — this fast onset is the unique property that lets it double as a reliever in MART/AIR regimens. Combined with budesonide or beclomethasone, it is the GINA preferred reliever and controller.

### BOARD TRAP

It is specifically formoterol's rapid onset (not just being a LABA) that allows reliever use — onset, not duration, is what distinguishes it from salmeterol for rescue.

## 10. Salmeterol — slow onset

### WHAT YOU SEE

An inhaler 'SALMETEROL' marked 'NOT for smart — slow onset'.

### WHAT IT ENCODES

Salmeterol is a long-acting LABA with SLOW onset (~10-20 min), so it provides maintenance bronchodilation but cannot relieve acute symptoms. It is used in fixed ICS-salmeterol combos for maintenance only.

### BOARD TRAP

Do not use salmeterol as a rescue/reliever inhaler — its slow onset means a patient in acute distress gets no timely relief; only formoterol is fast enough for rescue.

## 11. Montelukast (LTRA) add-on only

### WHAT YOU SEE

A 'LTRA montelukast' box marked 'add-on only — not monotherapy'.

### WHAT IT ENCODES

Leukotriene receptor antagonists (montelukast) block cysteinyl-leukotriene-mediated bronchoconstriction. They are alternative/add-on agents — inferior to ICS as monotherapy — but useful in aspirin-exacerbated respiratory disease (AERD), exercise-induced symptoms, and concomitant allergic rhinitis.

### BOARD TRAP

Montelukast is inferior to ICS as monotherapy and carries an FDA boxed warning for serious neuropsychiatric effects (depression, agitation, suicidality) — ask about mood changes before prescribing.

## 12. ACO overlap zone

### WHAT YOU SEE

Two workers 'ASTHMA' and 'COPD' at an 'ACO — overlap zone' with fluticasone/vilanterol.

### WHAT IT ENCODES

Asthma-COPD overlap (features of both: smoking history with reversible/eosinophilic disease) is managed with ICS-containing therapy (e.g., fluticasone/vilanterol), often triple therapy ICS-LABA-LAMA. The ICS is essential to cover the asthmatic/eosinophilic component.

### BOARD TRAP

In ACO never use LABA/LAMA without an ICS — unlike pure COPD where ICS may be omitted, the asthma component mandates inhaled steroid to prevent exacerbations and death.